Equilibrium balances components

LEARNING TARGET I can find a missing force when acceleration is zero.

SECTION A KEY VOCABULARY

Equilibrium	Zero a: forces sum to zero on each axis.
Balance	Vertical parts support the weight.
Symmetry	Equal angles and loading: equal tensions.
Rest	$v = 0$ alone does not prove balance.

SECTION B KEY EQUATIONS + UNITS

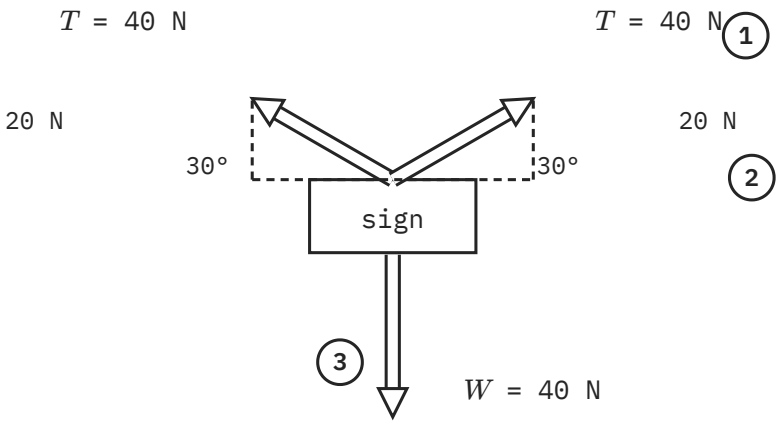
$$\Sigma F_x = \Sigma F_y = 0$$

$$2T \sin\theta = W$$

T force per rope, N
θ above horizontal

SECTION C COLOR IT · VERTICAL PARTS ADD

40 N sign; equal 30° ropes.



☐ **1 · ORANGE** Rope tensions

☐ **2 · ORANGE** Vertical parts: **20 N**

☐ **3 · ORANGE** Weight: **40 N**

SECTION D WORKED EXAMPLE

A **40 N** sign hangs at rest from two equal ropes, each **30°** above horizontal. **Find the tension in each rope.**

GIVEN	EQUATION	WORK	ANSWER
W = **40 N**; θ = **30°** equal tensions a = 0	x parts cancel. $2T \sin$ **30°** = **40 N**.	$2T(0.5)$ = **40 N** T = **40 N**	**40 N per rope** Each rope gives **20 N** upward.

SECTION E THE 2026-XJ THREAD

DAY 18 OF 22 · NEW TOOL: COMPONENT BALANCE

Zero net force would give XJ a constant velocity, not a guaranteed stop. Analysts must check each coordinate direction before treating its motion as steady.

Unit question: How fast is it moving, and where will it arrive?

UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced ☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else